

Problems

Status

Standings

斐波那契

Tags:

Time Limit: 3 s Memory Limit: 256 MB
Submission: 167 AC: 12 Score: 99.35

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Description

斐波那契数列的定义如下：
$$\begin{cases} F(x) = 0 & (x = 0) \\ F(x) = 1 & (x = 1) \\ F(x) = F(n - 1) + F(n - 2) & (x \geq 2) \end{cases}$$

即 $F=[1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, \dots]$
给出 n ，求 $(\sum_{i=1}^n F^2(i) + 2 * F(n-1) * F(n)) \bmod {10^9+7}$

Input

输入一个数 $n(1 \leq n \leq 10^{18})$

Output

输出结果

Samples

input	Copy
2	
output	Copy
4	

Author

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